

ATUL KUMAR

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LinkedIn

GitHub

Profile

ECE student with strong foundation in C++, DSA, and IoT development. Experienced in ESP32-based embedded systems and smart automation projects. Interested in software development and embedded systems. Quick learner with good communication and teamwork abilities.

Education

Godavari Institute of Engineering and Technology 2022 – 2026
B.Tech in Electronics and Communication Engineering (CGPA: 7.5)

Govt Higher Secondary School, Garh Dhamma 2022
12th (80%)

Experience

Intern – Embedded Systems / IoT Feb 2026 – May 2026
Genz Educate Wing

- Developed the Adaptive Smart Traffic Signal Controller project using ESP32 and FSM-based signal control logic.
- Worked on embedded systems, sensor integration, and real-time simulation using Wokwi Simulator.

Projects

Adaptive Smart Traffic Signal Controller
ESP32, Arduino IDE, Wokwi Simulator, OLED Display, LEDs, Push Buttons

- Designed an FSM-based intelligent traffic control system with adaptive signal timing based on traffic density.
- Implemented emergency vehicle priority and pedestrian crossing management with countdown display.
- Built OLED-based traffic monitoring and event logging system.

Smart Home Automation System
ESP32, Arduino IDE, PIR Sensor, LDR, DHT22, Relay Module, WiFi

- Developed an IoT-based smart home automation system using multiple sensors.
- Implemented automatic lighting, security monitoring, and temperature-based automation.

Green Energy Based Portable Charging Hub
Solar Energy, Power Electronics

- Designed a portable solar-powered charging hub focused on sustainable energy utilization.

Skills

- **Programming:** C++, Python (Basics)
- **Embedded / IoT:** ESP32, Arduino IDE, Wokwi Simulator, Relay Modules, PIR Sensor, LDR, DHT22
- **Core CS:** Data Structures and Algorithms (DSA)
- **Tools:** Git (Basics), Arduino IDE
- **Soft Skills:** Problem Solving, Communication, Teamwork, Adaptability
- **Languages:** English, Hindi, Telugu

Certifications & Achievements

- Blockchain and Applications – Great Learning
- Prompt Engineering – Great Learning
- Python Programming – Great Learning
- Cloud Fundamentals – Skill Camp
- Head Boy, Govt Higher Secondary School Garh Dhamma (2020–2021)
- PMSSS Scholarship Recipient
- Completed Higher Secondary Education with 80%